

Progress Test 3.C — Elec-S1

3.C · Operating point of a circuit
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Maximum power (1 mark)

A source has $E_{Th} = 12\text{ V}$ and $R_{Th} = 6.0\,\Omega$. Give the maximum power it can deliver to a resistive load.

[Empty grid for Exercise 1]

Marking

①	
0	1
☐	☐

Exercise 2. The operating point (5 marks)

A linear network, described by its Thévenin model (E_{Th}, R_{Th}), feeds a load resistor R_c . Define the operating point. Write down the characteristic of the source and that of the load, and deduce the analytical expressions of u^* and i^* .

[Empty grid for Exercise 2]

Marking

①		
0	1	2
☐	☐	☐

②		
0	1	2
☐	☐	☐

③	
0	1
☐	☐

(3 marks)

$\square \quad u^* \rightarrow 0 \text{ and } i^* \rightarrow E_{\text{Th}}/R_{\text{Th}}$	$\square \quad u^* \rightarrow E_{\text{Th}} \text{ and } i^* \rightarrow 0$
$\square \quad u^* \rightarrow E_{\text{Th}} \text{ and } i^* \rightarrow E_{\text{Th}}/R_{\text{Th}}$	$\square \quad u^* \rightarrow 0 \text{ and } i^* \rightarrow 0$

☐ 100% ☐ 75% ☐ 50% ☐ 25%

- ☐ matching, $R_c = R_{\text{Th}}$, to obtain the maximum power;
- ☐ a high efficiency, $R_c \gg R_{\text{Th}}$, to limit the losses;
- ☐ $R_c \ll R_{\text{Th}}$, to obtain the maximum current;
- ☐ a short circuit at the end of the line.

(5 marks)

$E_{\text{Th}} = 12\text{ V}$	$R_{\text{Th}} = 4.0\ \Omega$	$R_c = 8.0\ \Omega$
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①	
0	1

②	
0	1

③	
0	1

④	
0	1

⑤	
0	1

Unofficial practice material written from the Elec-St Part 1 lecture notes · not an official examination paper